Plugging into the Future:
An Exploration of Electricity Consumption Patterns

Team Members

MOHAMMED MAHABOOB ALISHA SHAIK

HEMANTH KALLEPALLI

SATISH VUJJURI

SUDHEEKSHA SANGANA

VELIGATLA PAVAN KUMAR

Project Final Report

Date 14 JULY 2026

1. INTRODUCTION

1.1 Project Overview

Electricity is one of the most essential resources for economic growth, industrial development, and daily human activities. Every day, electricity departments and energy organizations generate enormous amounts of consumption data from different states and regions. Analyzing this large volume of data manually is time-consuming and often fails to provide meaningful insights for planning and decision-making.

The **Electricity Consumption Analysis** project is developed to simplify electricity data analysis using modern Business Intelligence tools. The project utilizes **MySQL** for database management, **SQL** for analytical queries, **Tableau Desktop** for interactive dashboards, **Tableau Public** for online dashboard publishing, and **Flask** for deploying the dashboard as a web application.

The dashboard presents electricity consumption through interactive visualizations such as state-wise analysis, regional comparisons, monthly consumption trends, geographical maps, and top electricity-consuming states. Users can explore the dashboard using interactive filters and story points that clearly explain the analytical findings.

The completed dashboard is published on Tableau Public, while the project source code, documentation, SQL queries, and related files are maintained using GitHub. The project demonstrates how Business Intelligence tools can transform raw electricity data into meaningful visual insights that support effective planning and policy decisions.

1.2 Purpose

The primary purpose of this project is to analyze electricity consumption patterns across different states and regions using Business Intelligence techniques. The project converts raw electricity consumption data into interactive dashboards that help users identify trends, compare electricity usage, and understand regional demand patterns.

The project also aims to simplify electricity data analysis by presenting information through graphical visualizations rather than traditional reports. Government organizations, electricity boards, researchers, and policy makers can use these dashboards to make informed decisions regarding electricity generation, distribution, and resource planning.

Furthermore, this project demonstrates the practical application of data analytics, SQL, Tableau, and Flask in solving real-world energy management problems. It highlights the importance of data-driven decision-making in improving electricity planning, operational efficiency, and sustainable energy management.

2. IDEATION PHASE

Introduction

The Ideation Phase is the first stage of the project where different electricity-related problems were discussed and evaluated. Various project ideas were analyzed based on business value, implementation feasibility, availability of datasets, visualization potential, and real-world applicability.

Several ideas such as Smart Grid Monitoring, Renewable Energy Analysis, Electricity Demand Forecasting, Power Distribution Analysis, and Electricity Consumption Analysis were considered. After careful evaluation, **Electricity Consumption Analysis** was selected because it provides meaningful insights into electricity usage patterns and demonstrates the practical application of data analytics tools.

The ideation process included defining the problem statement, understanding stakeholder needs using the Empathy Map Canvas, and brainstorming multiple project ideas. This phase established a strong foundation for the Requirement Analysis, Project Design, and Development phases.

2.1 Problem Statement

Introduction

Electricity consumption data is generated continuously across different states and regions. However, traditional methods of analyzing this large volume of data are time-consuming, difficult to interpret, and often fail to provide actionable insights.

The proposed project addresses these challenges by developing an interactive Electricity Consumption Analysis system using MySQL, Tableau Desktop, Tableau Public, and Flask. The solution enables efficient data storage, analysis, visualization, and web-based dashboard access, allowing users to monitor electricity usage trends and support informed decision-making.

Problem Statement

Electricity departments and energy organizations collect large amounts of consumption data every day. Manual analysis of this data is inefficient and makes it difficult to identify consumption trends, compare electricity usage across regions, and support effective energy planning.

Therefore, there is a need for an interactive business intelligence solution that analyzes electricity consumption data and provides meaningful visualizations to assist decision-makers in understanding usage patterns, optimizing electricity management, and improving energy planning.

Existing Challenges

Challenge	**Impact**
Manual data analysis	Time-consuming and inefficient
Large electricity datasets	Difficult to interpret
Lack of visualization	Limited analytical insights
Traditional reporting	Delayed decision-making

Challenge	Impact
Regional comparison difficulties	Ineffective energy planning

Project Objectives

Objective ID	Objective
OBJ-1	Analyze electricity consumption data using Tableau Desktop
OBJ-2	Compare electricity usage across different states
OBJ-3	Study regional electricity consumption patterns
OBJ-4	Analyze monthly electricity consumption trends
OBJ-5	Develop interactive dashboards and Tableau Stories
OBJ-6	Publish dashboards through Tableau Public and Flask

Expected Outcomes

- Interactive electricity dashboards.
- Better understanding of electricity consumption patterns.
- Faster data analysis.
- Improved energy planning.
- Data-driven decision-making.
- Easy comparison between states and regions.

2.2 Empathy Map Canvas

Introduction

The **Empathy Map Canvas** is used to understand the needs, expectations, challenges, and goals of the stakeholders who use the Electricity Consumption Analysis system. It helps identify how users interact with electricity consumption data and what improvements can enhance their overall experience. By understanding the users' perspectives, the project was designed to provide meaningful visualizations, interactive dashboards, and accurate analytical insights.

The primary stakeholders include government organizations, electricity boards, utility companies, energy analysts, researchers, and policy makers. Their common objective is to monitor electricity consumption, identify usage patterns, and make informed decisions for effective energy management.

Empathy Map Canvas

Think and Feel

- Need accurate electricity consumption analysis.
- Want quick access to electricity usage information.
- Require interactive dashboards for easy analysis.
- Expect reliable and meaningful business insights.
- Need faster decision-making support.

Hear

- Government electricity policies.
- Recommendations from energy analysts.
- Reports from electricity boards.
- Consumer feedback regarding electricity demand.
- Industry best practices for energy management.

See

- Different electricity consumption across states.
- Regional variations in electricity demand.
- Monthly consumption trends.
- Geographic distribution of electricity usage.
- Interactive dashboards and visual reports.

Say and Do

- Analyze electricity consumption.
- Compare electricity usage across states.
- Monitor monthly electricity trends.
- Generate analytical reports.
- Share dashboards with stakeholders.
- Support planning and decision-making.

Pain

- Manual analysis requires significant time.
- Large datasets are difficult to understand.
- Traditional reports lack visualization.
- Identifying consumption trends is challenging.
- Delayed decision-making due to scattered information.

Gain

- Interactive Tableau dashboards.
- Faster and more accurate analysis.
- Easy comparison between states and regions.
- Better understanding of electricity demand.
- Improved planning and resource allocation.
- Data-driven decision-making.

Stakeholders

Stakeholder	**Responsibility**
Government Organizations	Monitor electricity consumption and plan policies
Electricity Boards	Manage electricity distribution and demand
Utility Companies	Analyze electricity usage and improve services
Energy Analysts	Perform data analysis and generate insights
Researchers	Study electricity consumption patterns
Business Users	Use dashboards for reporting and decision-making

---

Benefits of the Empathy Map

- Better understanding of stakeholder needs.
- Identification of major challenges in electricity data analysis.
- Development of user-friendly dashboards.
- Improved visualization and reporting.

- Enhanced decision-making through interactive analytics.
- Better user experience and accessibility.

Conclusion

The Empathy Map Canvas helped identify the expectations, challenges, and goals of the stakeholders involved in electricity consumption analysis. Understanding these requirements enabled the development of an interactive, user-friendly, and efficient analytics system. The insights gained from the empathy map guided the design of dashboards and visualizations that support better electricity management and informed decision-making.

2.3 Brainstorming

Introduction

Brainstorming is a creative process used to generate, evaluate, and select the most suitable project idea. During this phase, the team discussed multiple ideas related to data analytics and business intelligence. Each idea was analyzed based on real-world applicability, dataset availability, ease of implementation, visualization potential, and business value.

After evaluating all the proposed ideas, **Electricity Consumption Analysis** was selected because it effectively demonstrates the practical use of MySQL, SQL, Tableau, and Flask while addressing a real-world problem.

Brainstorming Ideas

S.No	Project Idea	Description
1	Electricity Consumption Analysis	Analyze electricity usage across different states and regions
2	Smart Grid Monitoring	Monitor power generation and distribution
3	Renewable Energy Analysis	Analyze renewable energy production and usage
4	Energy Demand Forecasting	Predict future electricity demand
5	Regional Electricity Monitoring	Compare electricity consumption across regions

---

Evaluation Criteria

Criteria	Description
Business Value	Solves a real-world problem
Dataset Availability	Suitable dataset is available

Criteria	Description
Visualization	Supports interactive dashboards
Feasibility	Easy to implement within project duration
Learning Opportunity	Demonstrates Business Intelligence tools

Selected Project

After evaluating all ideas, the team selected:

Plugging into the Future: An Exploration of Electricity Consumption Patterns

This project provides meaningful business insights through interactive dashboards and supports electricity planning using modern Business Intelligence tools.

Reasons for Selection

- Availability of a quality electricity consumption dataset.
- Supports interactive Tableau visualizations.
- Demonstrates SQL data analysis.
- Provides practical Business Intelligence experience.
- Helps understand electricity usage patterns.
- Supports better planning and decision-making.
- Suitable for real-world implementation.

Brainstorming Summary

Aspect	Outcome
Ideas Generated	5
Ideas Evaluated	5
Final Project Selected	Electricity Consumption Analysis
Development Tools	MySQL, SQL, Tableau, Flask
Expected Outcome	Interactive dashboard with business insights

Conclusion

The brainstorming process enabled the team to evaluate multiple project ideas and identify the most suitable solution. Based on feasibility, business relevance, and visualization capabilities, **Electricity Consumption Analysis** was selected as the final project. This decision established a strong foundation for the Requirement Analysis and Project Design phases while ensuring the project delivers meaningful analytical insights.

3. REQUIREMENT ANALYSIS

Introduction

The **Requirement Analysis** phase focuses on identifying the functional and non-functional requirements needed to develop the **Electricity Consumption Analysis** system. During this phase, user needs, system functionalities, data flow, and technology requirements were analyzed to ensure the project meets its objectives efficiently.

The system requirements were gathered by understanding the expectations of stakeholders such as government organizations, electricity boards, energy analysts, researchers, and business users. Based on these requirements, the project was designed to provide interactive dashboards, SQL-based analysis, geographical visualization, and web-based access to electricity consumption data.

This phase also defines the customer journey, solution requirements, data flow, and technology stack required for successful project implementation.

3.1 Customer Journey Map

Introduction

The **Customer Journey Map** illustrates the complete user experience while interacting with the Electricity Consumption Analysis system. It identifies each stage of user interaction, their goals, experiences, challenges, and opportunities for improvement.

Understanding the customer journey helps improve usability and ensures that the dashboard provides meaningful insights through an intuitive interface.

Customer Journey Map

Stage	**Experience**	**Interactions**	**Goals**	**Positive Moment**	**Negative Moment**	**Opportunity**
Access Dashboard	Opens the Flask application and accesses the Tableau dashboard.	Homepage loads and dashboard becomes available.	Quickly access electricity data.	Dashboard loads quickly.	Unsure where to begin.	Add a welcome page explaining dashboard features.
Review Electricity Statistics	Reviews state-wise and region-wise electricity charts.	Examines dashboards and visualizations.	Understand electricity consumption.	Easy comparison between states.	Large datasets may confuse new users.	Provide chart descriptions and tooltips.

Stage	Experience	Interactions	Goals	Positive Moment	Negative Moment	Opportunity
Analyze Trends	Uses filters to explore monthly electricity trends.	Selects regions, states, and dates.	Identify electricity usage patterns.	Interactive charts update instantly.	Multiple filters may overwhelm beginners.	Add a filter reset button and default selections.
Review Business Insights	Reads the Tableau Story and conclusions.	Reviews key findings and recommendations.	Understand electricity demand.	Story summarizes findings clearly.	Story requires updated data periodically.	Refresh dashboards regularly with new datasets.
Decision Making	Uses insights for planning and reporting.	Shares reports with stakeholders.	Improve electricity management.	Supports better planning.	Future demand prediction is unavailable.	Integrate AI-based electricity forecasting.

Customer Journey Summary

The customer journey demonstrates how users interact with the dashboard from accessing the application to making informed decisions. Interactive dashboards, visualizations, and stories provide an efficient and user-friendly experience while helping users understand electricity consumption patterns.

3.2 Solution Requirements

Introduction

The Electricity Consumption Analysis project requires both **functional** and **non-functional** requirements to ensure efficient data analysis, visualization, and system performance. These requirements define the expected behavior, quality, reliability, and usability of the system.

Functional Requirements

Requirement ID	Functional Requirement
FR-01	Import electricity consumption dataset into MySQL.
FR-02	Store electricity consumption data in the MySQL database.

Requirement ID	Functional Requirement
FR-03	Execute SQL queries for analytical calculations.
FR-04	Display state-wise electricity usage.
FR-05	Display region-wise electricity consumption.
FR-06	Analyze monthly electricity usage trends.
FR-07	Generate geographical maps using Tableau.
FR-08	Display the Top 10 electricity-consuming states.
FR-09	Create interactive Tableau dashboards.
FR-10	Create Tableau Story for business insights.
FR-11	Publish dashboards on Tableau Public.
FR-12	Embed Tableau dashboard into a Flask web application.

Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR-01	High system usability.
NFR-02	Fast SQL query execution.
NFR-03	Reliable dashboard performance.
NFR-04	Secure database management.
NFR-05	Easy dashboard accessibility.
NFR-06	High availability.
NFR-07	Scalable architecture.
NFR-08	Maintainable source code.
NFR-09	Interactive user interface.
NFR-10	Accurate data visualization.

Requirement Summary

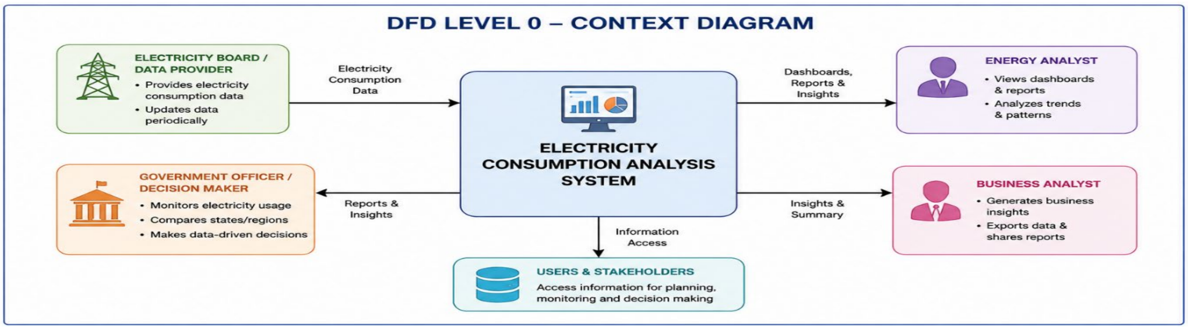
The identified requirements ensure that the Electricity Consumption Analysis system provides reliable data storage, efficient SQL analysis, interactive dashboards, web accessibility, and high-quality business insights.

3.3 Data Flow Diagram

Introduction

The **Data Flow Diagram (DFD)** illustrates how electricity consumption data flows through different components of the system. It explains the movement of data from collection to visualization and user interaction.

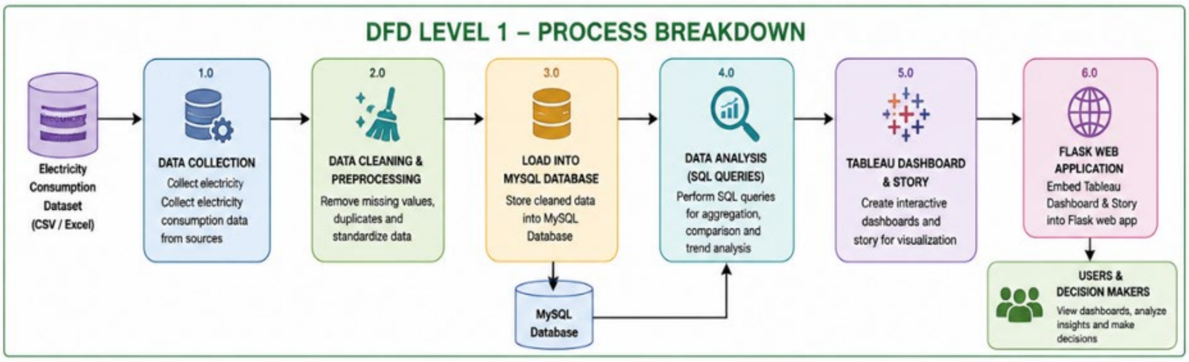
DFD Level 0



Description

The user provides the electricity consumption dataset, which is stored in the MySQL database. SQL queries process the data, Tableau generates interactive dashboards, and the results are displayed through the Flask web application.

DFD Level 1



Description

1. Electricity dataset is imported into MySQL.
2. SQL queries analyze the data.

3. Tableau Desktop creates charts and dashboards.
4. Dashboard is published on Tableau Public.
5. Flask embeds the dashboard.
6. End users access the dashboard and generate insights.

Data Flow Summary

The DFD demonstrates a structured flow of electricity consumption data from data collection to dashboard visualization and user interaction, ensuring efficient processing and accurate analysis.

3.4 Technology Stack

Introduction

The Electricity Consumption Analysis project integrates multiple technologies to support data storage, analysis, visualization, deployment, and documentation. Each technology plays a specific role in building a complete Business Intelligence solution.

Technology Stack

Component	Technology Used	Purpose
Dataset	CSV File	Stores electricity consumption data
Database	MySQL	Data storage and management
SQL Tool	MySQL Workbench	Query execution and database administration
Data Analysis	SQL	Data retrieval and aggregation
Visualization	Tableau Desktop	Create worksheets, dashboards, and stories
Dashboard Publishing	Tableau Public	Online dashboard sharing
Programming Language	Python 3.12.7	Backend development
Web Framework	Flask	Deploy Tableau dashboard through a web application
IDE	Visual Studio Code	Application development
Version Control	GitHub	Source code management
Documentation	Microsoft Word & PDF	Project documentation

Technology Integration

The project integrates MySQL, SQL, Tableau Desktop, Tableau Public, Flask, and GitHub into a single workflow. Data is stored in MySQL, analyzed using SQL, visualized through Tableau dashboards, published online, embedded in a Flask web application, and managed using GitHub for version control.

Requirement Analysis Conclusion

The Requirement Analysis phase successfully identified the functional requirements, non-functional requirements, customer journey, data flow, and technology stack required for the Electricity Consumption Analysis project. These requirements provide a strong foundation for designing and developing a reliable, scalable, and user-friendly analytics system that supports effective electricity management and informed decision-making.

4. PROJECT DESIGN

Introduction

The **Project Design Phase** focuses on designing an efficient solution for the **Electricity Consumption Analysis** project. During this phase, the overall architecture, workflow, proposed solution, and implementation strategy were defined. The design ensures that electricity consumption data is collected, processed, analyzed, visualized, and presented through an interactive dashboard.

The project integrates **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, **Flask**, and **Python** to develop a complete Business Intelligence solution. The designed architecture enables users to explore electricity consumption trends, compare state-wise and region-wise usage, and generate business insights for better decision-making.

4.1 Problem Solution Fit

Introduction

The Problem-Solution Fit identifies the relationship between the problems faced by users and the proposed solution offered by the Electricity Consumption Analysis system. It demonstrates how the project addresses real-world challenges through interactive visualization and data analytics.

Customer Segment

The project is designed for the following users:

- Government Organizations
- Electricity Boards
- Utility Companies
- Energy Analysts
- Researchers
- Business Users
- Students

Customer Problems

Users experience several challenges while analyzing electricity consumption data.

- Large electricity datasets
- Manual data analysis
- Difficulty identifying trends
- Lack of visualization

- Slow report generation
- Limited decision support

Existing Solutions

Current methods include:

- Microsoft Excel
- Manual Reports
- Static Graphs
- SQL Reports
- Spreadsheet Analysis

These methods provide limited visualization and are not interactive.

Proposed Solution

The proposed solution is an **interactive Electricity Consumption Analysis Dashboard** developed using MySQL, Tableau Desktop, Tableau Public, and Flask.

The solution enables users to:

- Analyze electricity consumption.
- Compare electricity usage across states.
- Compare regional electricity demand.
- Monitor monthly electricity trends.
- View geographical electricity distribution.
- Generate business insights.
- Access dashboards through a web browser.

Problem Solution Fit Table

Problem	**Proposed Solution**	**Expected Benefit**
Manual analysis	Interactive Tableau Dashboard	Faster analysis
Large datasets	MySQL Database	Efficient storage
Difficult comparisons	Charts and Maps	Easy comparison
Static reports	Interactive Dashboard	Better visualization

Problem	Proposed Solution	Expected Benefit
Limited accessibility	Flask Web Application	Easy web access
Decision making	Business Insights	Better planning

Expected Benefits

- Faster analysis
- Interactive dashboards
- Better visualization
- Accurate SQL analysis
- Improved planning
- Better resource allocation
- User-friendly interface
- Web accessibility

Conclusion

The Problem-Solution Fit demonstrates that the proposed Electricity Consumption Analysis system effectively addresses the limitations of traditional electricity reporting methods. The integration of MySQL, Tableau, and Flask provides an efficient, scalable, and interactive solution for electricity consumption analysis.

4.2 Proposed Solution

Introduction

The proposed solution combines modern Business Intelligence technologies to transform raw electricity consumption data into interactive dashboards and analytical reports.

The solution begins with data collection, followed by database storage, SQL analysis, visualization using Tableau, dashboard publishing through Tableau Public, and deployment using Flask.

Proposed Solution Overview

The Electricity Consumption Analysis system performs the following activities:

- Collect electricity consumption dataset.
- Clean and validate data.
- Store the dataset in MySQL.

- Execute SQL analytical queries.
- Create Tableau worksheets.
- Develop interactive dashboards.
- Create Tableau Story.
- Publish dashboards on Tableau Public.
- Integrate Tableau dashboard into Flask.
- Deploy through a web application.

Features of Proposed Solution

Feature	Description
Database Management	Store electricity data using MySQL
SQL Analysis	Calculate usage statistics
Interactive Dashboard	Visualize electricity consumption
Geographic Map	Display state-wise electricity usage
Tableau Story	Present business insights
Flask Application	Web-based dashboard access
Tableau Public	Online dashboard publishing

Workflow

The proposed solution follows this workflow:

```
Electricity Dataset (CSV)
        |
        ▼
Data Cleaning & Validation
        |
        ▼
MySQL Database
        |
        ▼
SQL Query Analysis
        |
        ▼
Tableau Desktop
        |
```

▼
Interactive Worksheets
|
▼
Dashboard Development
|
▼
Story Creation
|
▼
Tableau Public
|
▼
Flask Web Application
|
▼
End Users

Advantages of Proposed Solution

- Interactive visualizations
- Faster analysis
- Accurate reporting
- Better decision making
- Geographic analysis
- Easy dashboard access
- Scalable architecture
- Improved user experience

Conclusion

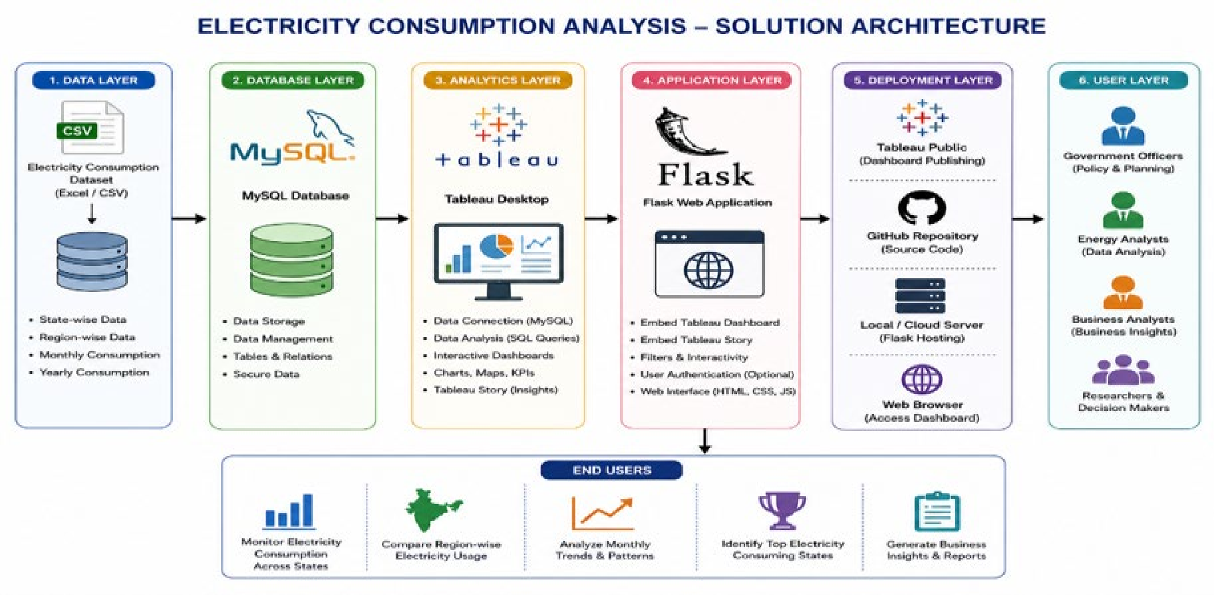
The proposed solution provides an efficient platform for analyzing electricity consumption data using MySQL, Tableau, and Flask. The integration of these technologies enables interactive dashboards, meaningful business insights, and easy web accessibility.

4.3 Solution Architecture

Introduction

The Solution Architecture defines the structure of the Electricity Consumption Analysis system. It explains how various technologies interact to process electricity consumption data from collection to visualization and web deployment.

Solution Architecture



Architecture Components

Data Layer

- Electricity Consumption Dataset
- Data Validation
- Data Cleaning

Database Layer

- MySQL Database
- SQL Queries
- Data Storage

Analytics Layer

- Tableau Desktop
- Worksheets
- Dashboard
- Story

Deployment Layer

- Tableau Public
- Flask Web Application

User Layer

- Government Organizations
- Energy Analysts
- Utility Companies
- Researchers
- Business Users

Working Process

1. Collect electricity consumption dataset.
2. Validate and clean the dataset.
3. Store data in MySQL.
4. Execute SQL queries.
5. Create Tableau worksheets.
6. Develop dashboards.
7. Create Tableau Story.
8. Publish dashboard on Tableau Public.
9. Embed dashboard into Flask.
10. Users access interactive dashboards.

Architecture Advantages

- Efficient data management
- Interactive visualization
- Faster analysis
- Better scalability
- Easy deployment
- Web accessibility
- Reliable decision support

Project Design Summary

The Project Design Phase successfully defined the architecture, workflow, and implementation strategy for the Electricity Consumption Analysis system. The integration of MySQL, SQL, Tableau, Tableau Public, and Flask provides a scalable and efficient Business Intelligence solution for electricity consumption analysis.

Conclusion

The Project Design Phase established a strong foundation for implementing the Electricity Consumption Analysis project. The proposed architecture ensures efficient data processing, interactive visualization, and easy accessibility. The system supports informed decision-making and provides a scalable platform for future enhancements in electricity data analytics.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Introduction

Project Planning is one of the most important phases of the Software Development Life Cycle (SDLC). It defines the roadmap for completing the project successfully by identifying project activities, allocating resources, estimating timelines, identifying risks, and monitoring progress.

For the **Electricity Consumption Analysis** project, a structured planning approach was followed to ensure systematic execution from project ideation to final deployment. Each activity was carefully scheduled to complete the project within the planned timeframe while maintaining quality and achieving the project objectives.

The project planning process included dataset collection, database development, SQL query implementation, Tableau dashboard development, Flask web application integration, testing, documentation, GitHub repository management, and final project submission.

Project Planning Objectives

The major objectives of project planning are:

- Define the scope of the project.
- Establish a realistic development schedule.
- Allocate resources effectively.
- Identify potential project risks.
- Monitor project progress.
- Ensure timely project completion.
- Deliver a high-quality Business Intelligence solution.

Project Planning Overview

The project was divided into multiple development phases to ensure smooth implementation.

The planning process included:

- Project topic selection
- Dataset collection
- Database development
- SQL analysis
- Tableau dashboard creation
- Tableau Story development

- Dashboard publishing
- Flask integration
- Testing
- Documentation
- GitHub repository creation
- Final submission

A structured weekly schedule was prepared to monitor the completion of each activity.

Project Schedule

Week	Activity
Week 1	Project topic selection and problem identification
Week 2	Dataset collection, cleaning, and preprocessing
Week 3	MySQL database creation and SQL query implementation
Week 4	Tableau worksheet development
Week 5	Interactive dashboard and Tableau Story creation
Week 6	Tableau Public publishing and Flask web application integration
Week 7	Testing, GitHub repository creation, and validation
Week 8	Documentation, report preparation, and final project submission

---

Project Planning Components

The planning phase consists of the following major components.

1. Requirement Analysis

- Study stakeholder requirements.
- Analyze project objectives.
- Define functional requirements.
- Define non-functional requirements.

---

2. Database Planning

- Design MySQL database.

- Import electricity dataset.
- Execute SQL queries.
- Validate data.

---

3. Dashboard Planning

- Design Tableau worksheets.
- Create interactive dashboard.
- Develop Tableau Story.
- Add filters and formatting.

---

4. Deployment Planning

- Publish dashboard on Tableau Public.
- Integrate dashboard with Flask.
- Test web application.
- Verify browser compatibility.

---

5. Documentation Planning

- Prepare APSCHE reports.
- Maintain GitHub repository.
- Capture screenshots.
- Prepare final presentation.

Project Milestones

Milestone	Status
Project Topic Finalized	Completed
Dataset Collected	Completed
Database Created	Completed
SQL Queries Executed	Completed
Tableau Worksheets Developed	Completed

Milestone	Status
Interactive Dashboard Created	Completed
Tableau Story Completed	Completed
Dashboard Published	Completed
Flask Application Developed	Completed
GitHub Repository Created	Completed
Documentation Completed	Completed

Resource Planning

The following software tools were used during project development.

Resource	Purpose
MySQL Workbench	Database management
SQL	Data analysis
Tableau Desktop	Dashboard development
Tableau Public	Dashboard publishing
Flask	Web application
Python	Backend programming
Visual Studio Code	Code development
GitHub	Version control
Microsoft Word	Documentation

Risk Management

During project development, several risks were identified.

Risk	Impact	Mitigation
Missing dataset values	Incorrect analysis	Data preprocessing
SQL query errors	Wrong calculations	Query validation

Risk	Impact	Mitigation
Tableau connection issues	Dashboard failure	Verify database connection
Dashboard publishing issues	Deployment delay	Publish verification
Flask integration issues	Application failure	Local testing
Documentation delays	Late submission	Weekly progress monitoring

---

Deliverables

The project planning phase identified the following deliverables.

- Cleaned Electricity Consumption Dataset
- MySQL Database
- SQL Query File
- Tableau Workbook
- Interactive Dashboard
- Tableau Story
- Tableau Public Dashboard
- Flask Web Application
- GitHub Repository
- APSCHE Documentation
- Final Project Report

Expected Outcome

The Project Planning Phase provides a structured roadmap for implementing the Electricity Consumption Analysis project. Proper scheduling, resource allocation, and risk management ensure timely completion of all development activities while maintaining project quality.

The planning process supports efficient project execution and establishes a strong foundation for successful implementation.

Project Planning Summary

Activity	Outcome
Planning	Successful

Activity	Outcome
Scheduling	Completed
Resource Allocation	Completed
Risk Analysis	Completed
Deliverables Identified	Completed
Timeline Management	Completed

Conclusion

The **Project Planning & Scheduling** phase established a well-organized roadmap for successfully implementing the **Electricity Consumption Analysis** project. Through systematic scheduling, resource planning, milestone tracking, and risk management, the project activities were completed efficiently within the planned timeline. This phase ensured proper coordination among all development tasks and laid a strong foundation for successful implementation, testing, deployment, and documentation of the project.

6. PROJECT DEVELOPMENT PHASE

Introduction

The **Project Development Phase** is the implementation stage of the **Electricity Consumption Analysis** project. During this phase, the planned solution was transformed into a functional Business Intelligence application by integrating **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, and **Flask**.

The development process involved collecting and preprocessing the electricity consumption dataset, creating the MySQL database, performing SQL analysis, designing interactive Tableau dashboards and stories, publishing the dashboard through Tableau Public, integrating it into a Flask web application, and maintaining the project using GitHub.

This phase ensured that the system was developed systematically while meeting all functional and non-functional requirements identified during the Requirement Analysis phase.

6.1 Dataset Collection

Introduction

The first step of the development process involved collecting a reliable electricity consumption dataset. The dataset contains electricity usage information from different states and regions of India and serves as the primary source for analysis.

The dataset was carefully examined before importing it into the MySQL database.

Dataset Information

- Dataset Format: CSV (.csv)
- Total Records: 16,599
- Database Name: electricity
- Table Name: consumption

Dataset Attributes

The dataset contains the following information:

- State
- Region
- Latitude
- Longitude
- Date
- Electricity Consumption
- Month
- Year

Data Preprocessing

Before analysis, the dataset was cleaned by:

- Removing duplicate records
- Handling missing values
- Validating electricity consumption values
- Standardizing data format
- Checking data consistency

These preprocessing steps improved the accuracy and reliability of the analytical results.

6.2 Database Development

Introduction

The electricity dataset was imported into a **MySQL** database to facilitate efficient storage, retrieval, and analysis.

A database named **electricity** was created, and the dataset was imported into the **consumption** table.

Database Activities

- Database Creation
- Table Creation
- CSV Import
- Data Validation
- SQL Query Execution

SQL Analysis

Several SQL queries were developed to analyze electricity consumption.

These include:

- Total Electricity Consumption
- Average Electricity Consumption
- Maximum Electricity Consumption
- Minimum Electricity Consumption
- State-wise Electricity Consumption
- Region-wise Electricity Consumption
- Top 10 Electricity Consumers
- Monthly Electricity Consumption

The SQL queries generated accurate analytical results used in Tableau dashboards.

6.3 Tableau Dashboard Development

Introduction

The processed MySQL data was connected to Tableau Desktop for visualization.

Interactive worksheets were created to present electricity consumption from different perspectives.

Worksheets Developed

- State-wise Electricity Consumption
- Region-wise Electricity Consumption
- Monthly Electricity Usage Trend
- State-wise Electricity Consumption Map
- Top 10 Electricity Consumers

Each worksheet was designed using appropriate charts, labels, filters, and formatting to improve readability.

6.4 Dashboard Development

After developing individual worksheets, all visualizations were combined into a single interactive dashboard.

Dashboard Name:- Electricity Consumption Dashboard

Dashboard Features

- KPI Summary
- State-wise Comparison
- Region-wise Analysis
- Monthly Trends
- Geographic Visualization
- Top 10 Electricity Consumers
- Interactive Filters
- Dynamic Navigation

The dashboard enables users to explore electricity consumption through multiple perspectives.

6.5 Tableau Story Development

Introduction

A Tableau Story was created to present the analysis in a structured sequence.

Story Points

1. Project Overview
2. Region-wise Electricity Consumption
3. State-wise Electricity Consumption
4. Monthly Electricity Usage Trend
5. State-wise Electricity Consumption Map
6. Top 10 Electricity Consumers
7. Business Insights & Conclusion

The story explains the analytical process and highlights key findings for stakeholders.

6.6 Tableau Public Publishing

After completing the dashboard and story, the project was published on **Tableau Public**.

Publishing the dashboard allows users to:

- Access the dashboard online
- Interact with filters
- Explore visualizations
- Share analytical insights

The Tableau Public dashboard can be accessed through a web browser.

6.7 Flask Web Application Development

Introduction

A web application was developed using **Python Flask**.

The published Tableau dashboard was embedded into the Flask application using HTML.

Features

- User-friendly interface
- Dashboard embedding
- Easy navigation
- Browser accessibility

- Local deployment

The application can be accessed through:

http://127.0.0.1:5000

6.8 GitHub Repository

The entire project was uploaded to GitHub for version control and documentation.

Repository Contents

- Dataset
- SQL Queries
- MySQL Scripts
- Tableau Workbook
- Dashboard
- Tableau Story
- Flask Application
- Source Code
- Reports
- Screenshots
- README.md
- requirements.txt
- . gitignore
- LICENSE

GitHub enables secure storage, collaboration, and version tracking.

6.9 Development Workflow

The project follows the workflow below.

Electricity Dataset (CSV)
│
▼
Data Cleaning & Validation
│
▼
MySQL Database
│
▼

```
SQL Query Processing
        |
        ▼
Tableau Desktop
        |
        ▼
Worksheets
        |
        ▼
Dashboard
        |
        ▼
Tableau Story
        |
        ▼
Tableau Public
        |
        ▼
Flask Web Application
        |
        ▼
GitHub Repository
        |
        ▼
End Users
```

6.10 Project Outputs

The completed project successfully delivers:

- Interactive Electricity Dashboard
- SQL-based Analysis
- State-wise Electricity Analysis
- Region-wise Comparison
- Monthly Trend Analysis
- Geographic Visualization
- Top 10 Electricity Consumers
- Tableau Story
- Tableau Public Dashboard
- Flask Web Application
- GitHub Repository

6.11 Development Summary

Component	**Status**
Dataset Collection	Completed
Database Development	Completed
SQL Queries	Completed
Tableau Worksheets	Completed
Dashboard Development	Completed
Story Development	Completed
Tableau Public Publishing	Completed
Flask Integration	Completed
GitHub Repository	Completed
Documentation	Completed

Conclusion

The **Project Development Phase** successfully implemented the **Electricity Consumption Analysis** system by integrating **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, **Flask**, and **GitHub**. The project transformed raw electricity consumption data into interactive dashboards, meaningful visualizations, and business insights. The developed solution enables efficient electricity data analysis, supports informed decision-making, and provides a scalable platform for future enhancements such as real-time monitoring and AI-based electricity demand forecasting.

7. FUNCTIONAL & PERFORMANCE TESTING

Introduction

The **Functional and Performance Testing Phase** was conducted to verify that the **Electricity Consumption Analysis** project meets all functional and non-functional requirements. Testing ensures that the MySQL database, SQL queries, Tableau dashboards, Tableau Story, Tableau Public dashboard, Flask web application, and GitHub repository function correctly under normal operating conditions.

The testing process evaluated the system for **accuracy, performance, usability, compatibility, and reliability**. Different test cases were executed to ensure that all project components produced correct results and provided a smooth user experience.

The successful completion of this phase confirms that the developed system is stable, reliable, and ready for practical use.

7.1 Accuracy Testing

Introduction

Accuracy Testing was conducted to verify that the SQL query results, Tableau dashboards, and visualizations accurately represent the electricity consumption dataset stored in the MySQL database.

The analytical results generated through SQL queries were compared with the original dataset to ensure correctness.

Accuracy Testing Activities

- Verification of imported dataset
- SQL query validation
- Dashboard value verification
- Chart accuracy validation
- KPI verification
- Tableau Story verification

Accuracy Testing Results

- State-wise electricity consumption matched the dataset.
- Region-wise analysis produced accurate results.
- Monthly electricity trends were correctly displayed.
- Top 10 electricity-consuming states matched SQL outputs.
- Dashboard calculations were accurate.
- Tableau Story presented correct analytical insights.

The testing confirmed that all analytical results are reliable and consistent.

7.2 Performance Testing

Introduction

Performance Testing evaluated the responsiveness and efficiency of the Electricity Consumption Analysis system.

The project was tested using the complete dataset containing **16,599 records**.

Performance Testing Parameters

- SQL query execution
- Dashboard loading speed
- Filter responsiveness
- Story navigation
- Flask application response
- Tableau Public performance

Performance Testing Results

- SQL queries executed efficiently.
- Dashboard loaded quickly.
- Interactive filters responded instantly.
- Story navigation was smooth.
- Flask application displayed dashboards successfully.
- Tableau Public dashboard loaded without noticeable delay.

The system maintained stable performance throughout testing.

7.3 Usability Testing

Introduction

Usability Testing ensured that users could easily understand and interact with the dashboard.

The dashboard was evaluated based on simplicity, navigation, readability, and user experience.

Usability Checklist

- Easy navigation
- Interactive filters
- Readable charts
- Proper labels

- Dashboard layout
- Story sequence
- Business insights

Results

Users successfully:

- Compared electricity consumption.
- Applied dashboard filters.
- Understood geographical maps.
- Identified electricity trends.
- Reviewed business insights.

The dashboard was intuitive and user-friendly.

7.4 Compatibility Testing

Introduction

Compatibility Testing verified that the project works correctly across different software environments and browsers.

Tested Platforms

Platform	Status
Windows 11	Passed
Google Chrome	Passed
Microsoft Edge	Passed
MySQL Workbench	Passed
Tableau Desktop	Passed
Tableau Public	Passed
Visual Studio Code	Passed
Flask Local Server	Passed

The project functioned successfully on all tested platforms.

7.5 Security Testing

Introduction

Security Testing ensured that project files and data remained protected during development.

Security Measures

- Local MySQL database.
- Version control using GitHub.
- Backup of Tableau workbooks.
- Secure project documentation.
- Controlled access to source code.

No major security issues were identified during testing.

Performance Testing Summary

Testing Type	Objective	Result
Accuracy Testing	Verify dashboard calculations	Passed
Performance Testing	Evaluate dashboard responsiveness	Passed
Usability Testing	Verify user friendliness	Passed
Compatibility Testing	Test across platforms	Passed
Security Testing	Protect project resources	Passed

Test Cases

Test Case	Expected Result	Actual Result	Status
Import CSV into MySQL	Data imported successfully	16,599 records imported	✓ Passed
Execute SQL Queries	Correct analytical output	Successfully executed	✓ Passed
Connect Tableau with MySQL	Successful connection	Connected successfully	✓ Passed
Create Worksheets	Worksheets generated	Successfully created	✓ Passed
Create Dashboard	Interactive dashboard	Successfully created	✓ Passed
Create Tableau Story	Story displays correctly	Successfully created	✓ Passed

Test Case	Expected Result	Actual Result	Status
Publish Tableau Dashboard	Dashboard available online	Successfully published	✓ Passed
Flask Integration	Dashboard displayed in browser	Successfully displayed	✓ Passed
GitHub Repository	Files uploaded	Successfully uploaded	✓ Passed

---

Test Results

The testing process confirmed that all major components of the Electricity Consumption Analysis project operated successfully. SQL queries generated accurate analytical results, Tableau dashboards displayed correct visualizations, Tableau Stories presented business insights effectively, and the Flask web application embedded the dashboard without issues.

Interactive filters, maps, charts, and dashboard navigation functioned smoothly, ensuring a positive user experience. Compatibility testing verified that the system worked consistently across supported platforms and browsers.

Overall Testing Result

The **Electricity Consumption Analysis** project successfully passed all functional and performance tests. The system demonstrated high accuracy, efficient performance, good usability, reliable compatibility, and stable operation. All planned functionalities were successfully validated, confirming that the project is ready for deployment and practical use.

Testing Summary

Component	Status
MySQL Database	✓ Completed
SQL Queries	✓ Completed
Tableau Worksheets	✓ Completed
Dashboard	✓ Completed
Tableau Story	✓ Completed
Tableau Public	✓ Completed
Flask Application	✓ Completed
GitHub Repository	✓ Completed

Conclusion

The **Functional and Performance Testing Phase** verified that the Electricity Consumption Analysis system meets the required standards for accuracy, performance, usability, compatibility, and security. All project components functioned correctly and produced reliable analytical results. The successful completion of testing confirms that the system is stable, user-friendly, and capable of supporting efficient electricity consumption analysis and informed decision-making.

8. RESULTS

Introduction

The **Results** section presents the outcomes obtained after successfully implementing the **Electricity Consumption Analysis** project. The project transformed raw electricity consumption data into meaningful visual insights using **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, and **Flask**.

Interactive dashboards, geographical maps, and Tableau Stories enabled users to analyze electricity consumption across different states and regions. The results demonstrate the effectiveness of Business Intelligence techniques in supporting electricity planning and decision-making.

8.1 SQL Query Results

The electricity consumption dataset was successfully imported into the MySQL database, and SQL queries were executed to analyze the data.

SQL Analysis Performed

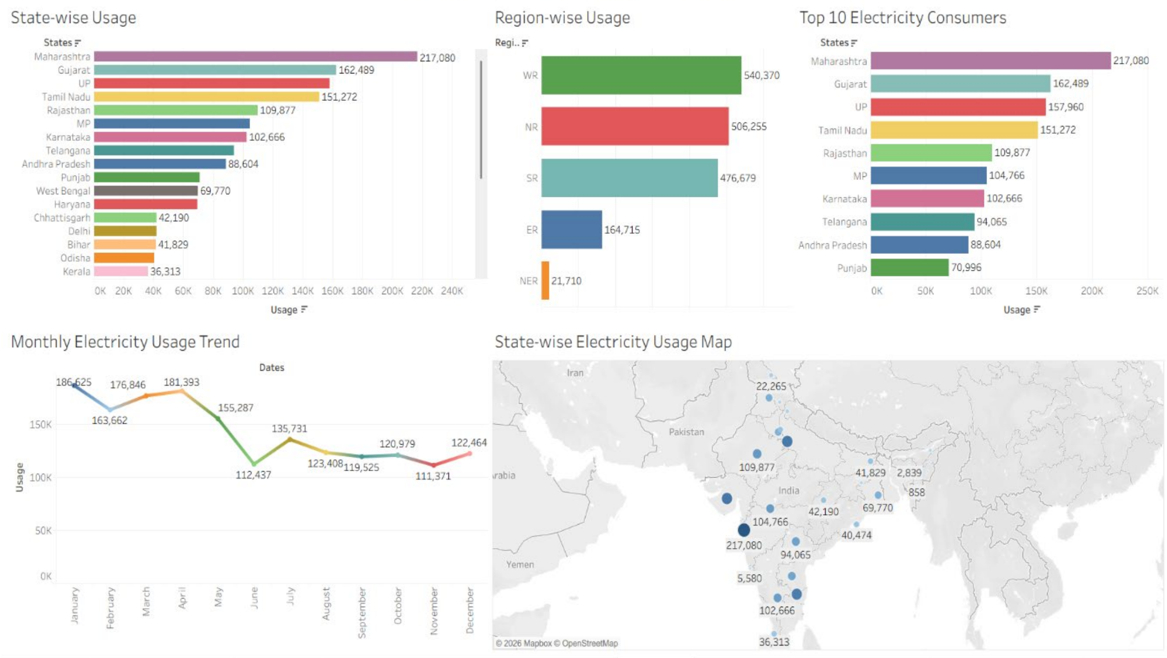
- Total Electricity Consumption
- Average Electricity Consumption
- Maximum Electricity Consumption
- Minimum Electricity Consumption
- State-wise Electricity Consumption
- Region-wise Electricity Consumption
- Monthly Electricity Consumption
- Top 10 Electricity Consuming States

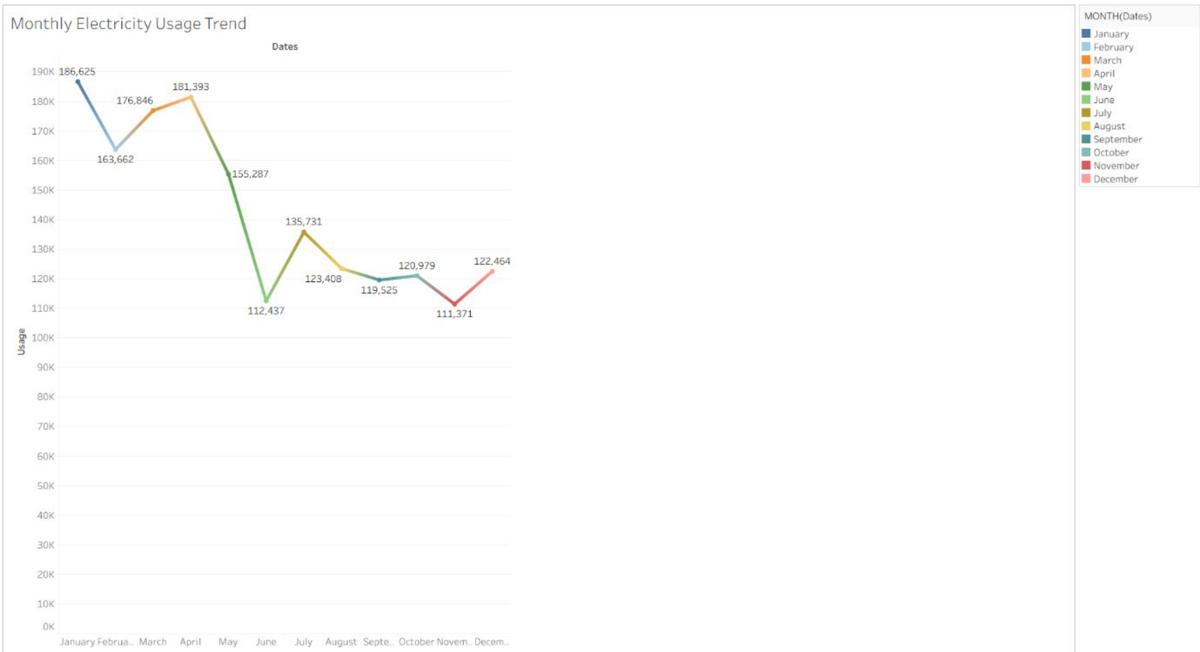
8.2 Tableau Worksheets

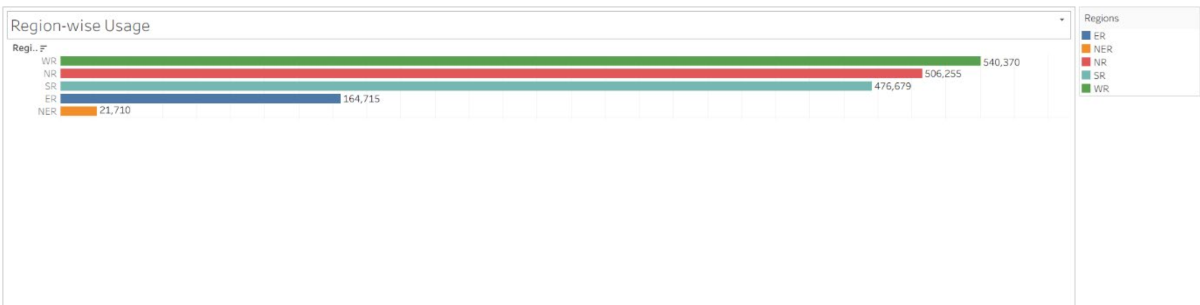
The following worksheets were successfully created in Tableau Desktop.

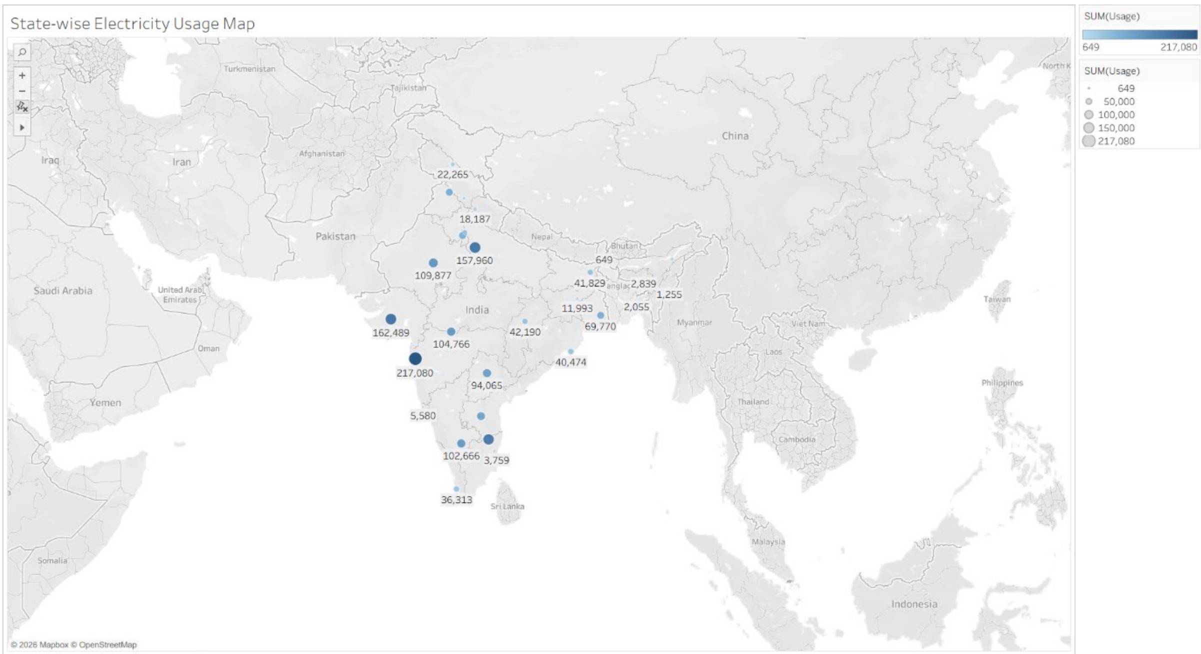
Worksheet	Purpose
State-wise Electricity Usage	Compare electricity usage across states
Region-wise Electricity Usage	Compare regional electricity consumption
Monthly Electricity Trend	Analyze monthly usage patterns
State-wise Electricity Map	Display electricity consumption geographically
Top 10 Electricity Consumers	Identify highest electricity-consuming states

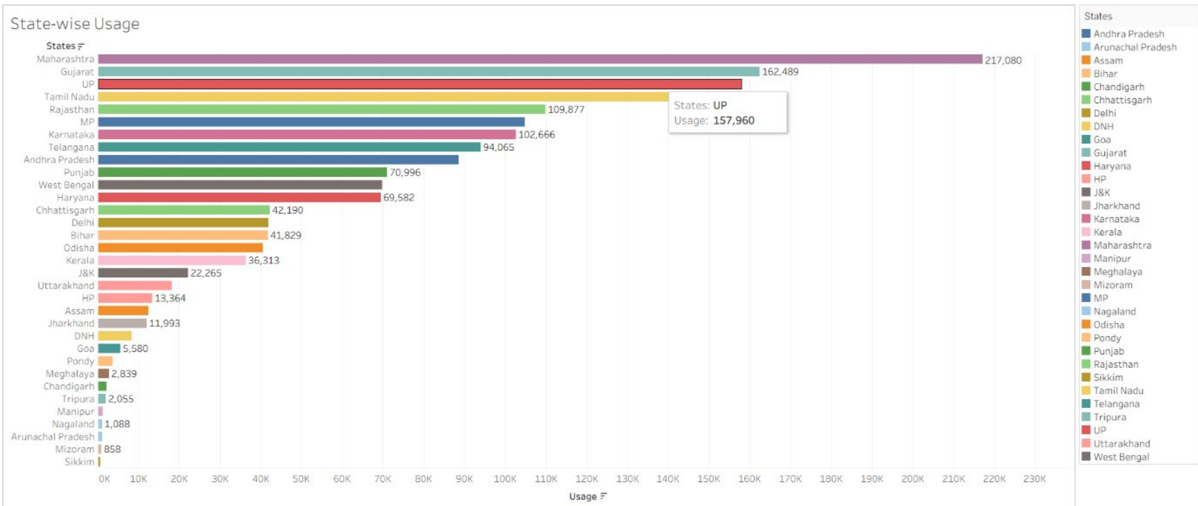
Electricity Consumption Analysis Dashboard

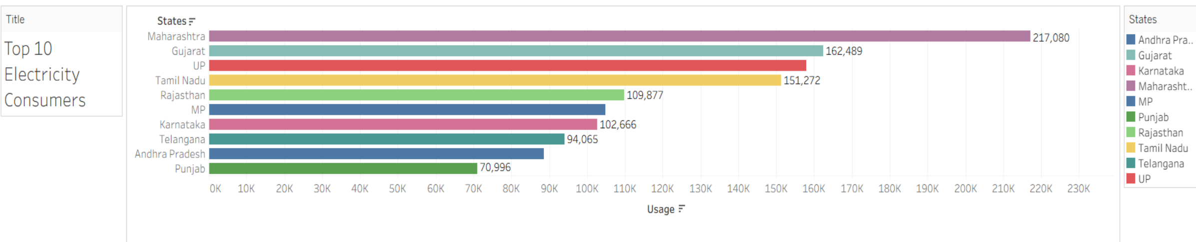












8.3 Interactive Dashboard

The individual worksheets were combined into a single interactive dashboard.

Dashboard Features

- KPI Summary

- State-wise Analysis
- Region-wise Analysis
- Monthly Trends
- Geographic Map
- Top 10 Electricity Consumers
- Interactive Filters

The dashboard enables users to explore electricity consumption from different perspectives and supports efficient analysis.

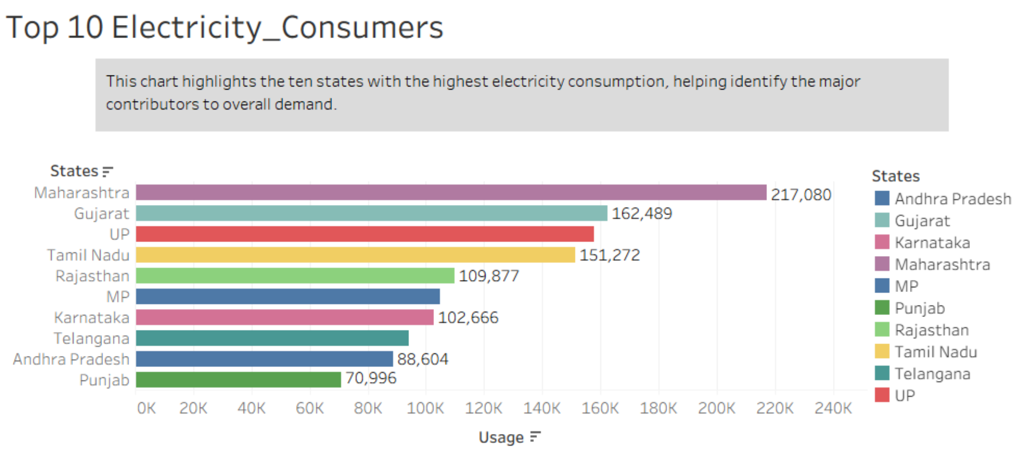
8.4 Tableau Story

The Tableau Story presents the analysis in a structured format.

Story Pages

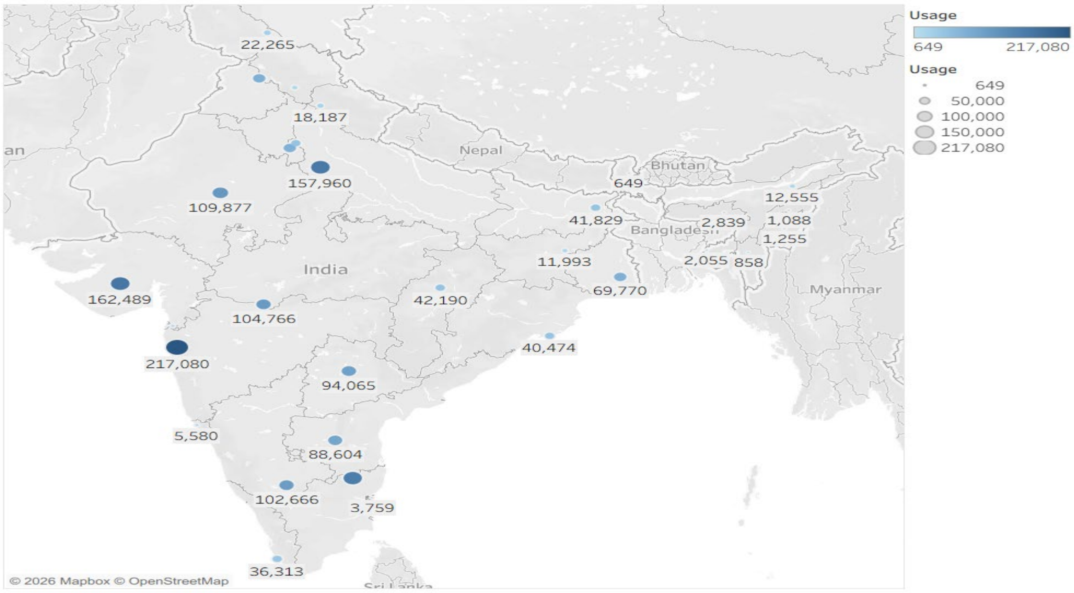
1. Project Overview
2. Region-wise Electricity Usage
3. State-wise Electricity Usage
4. Monthly Electricity Trend
5. State-wise Electricity Map
6. Top 10 Electricity Consumers
7. Business Insights & Conclusion

The story helps users understand the analytical findings step by step.



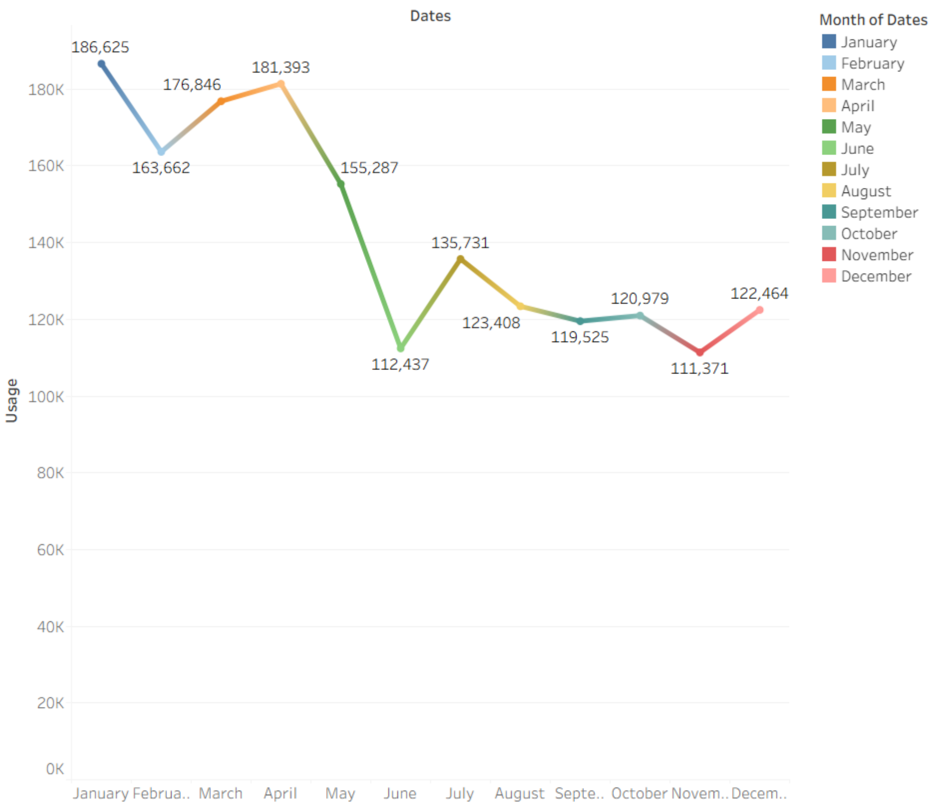
State-wise Electricity Usage_Map

The map highlights the geographical distribution of electricity consumption across India, making it easy to identify high and low usage states.



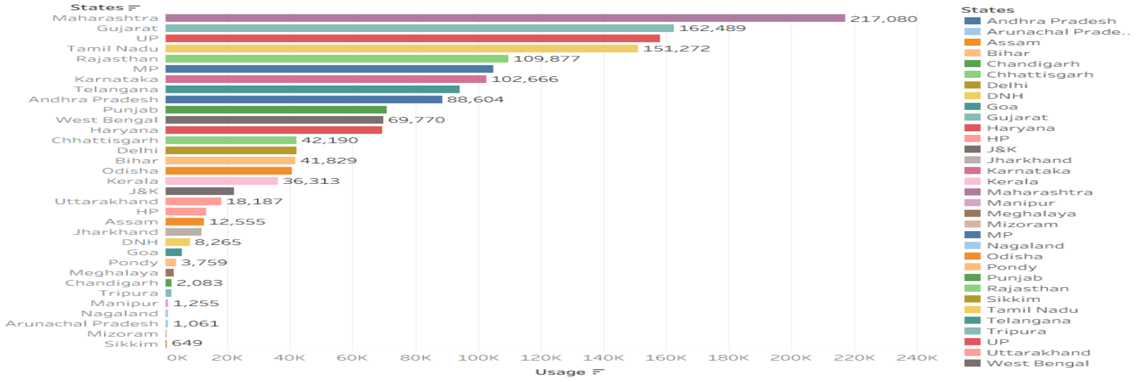
Monthly Electricity Usage_Trend

The monthly trend shows seasonal variations in electricity consumption, helping identify peak and low-demand periods throughout the year.



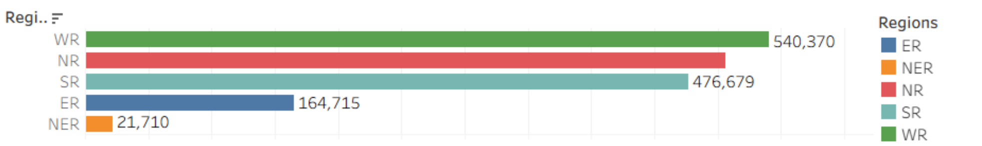
State-wise Electricity Usage

This visualization compares electricity consumption across different states. Maharashtra has the highest electricity usage, while smaller states consume comparatively less.



Region-wise Electricity Usage

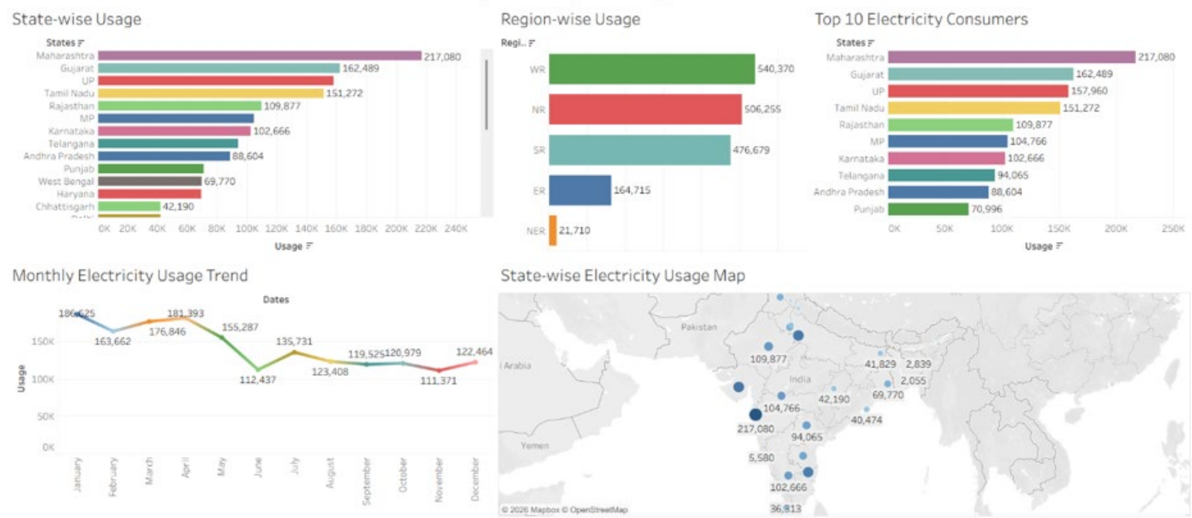
The western region records the highest electricity consumption, followed by the northern and southern regions, while the north-eastern region has the lowest usage.



Project Overview

This dashboard provides an overview of electricity consumption across Indian states, regions, monthly trends, and geographical distribution.

Electricity Consumption Analysis Dashboard

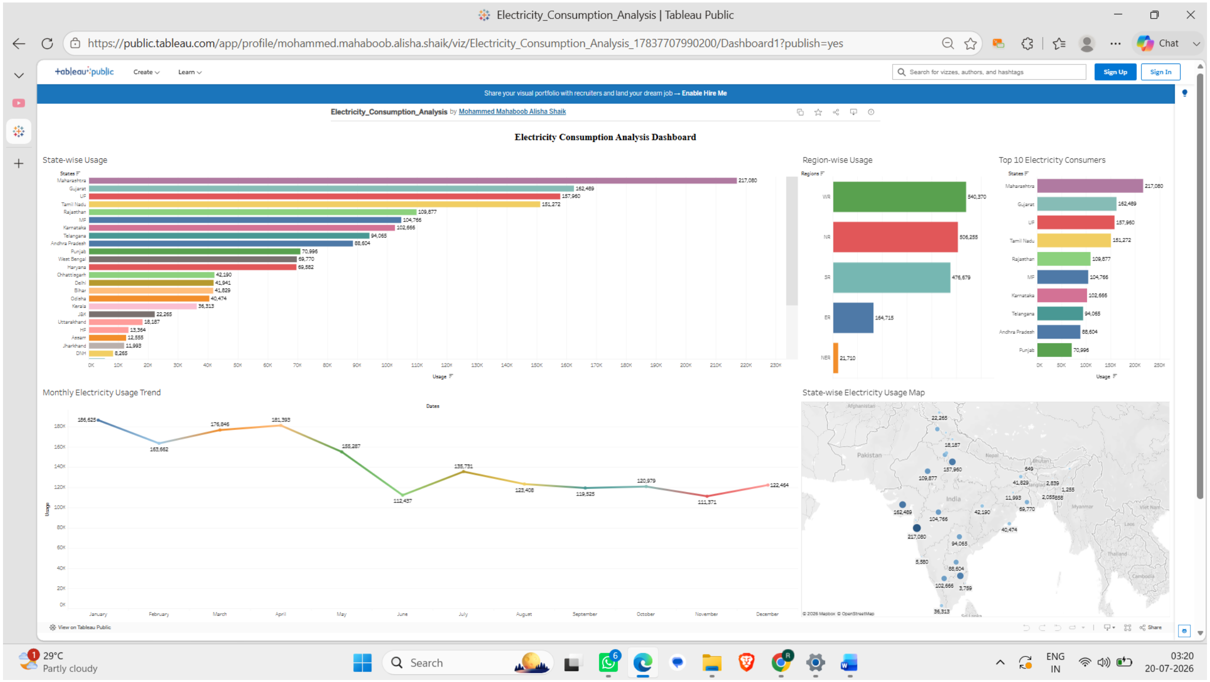


8.5 Tableau Public

The completed dashboard was successfully published on **Tableau Public**, allowing users to access and interact with the dashboard online.

Benefits

- Online accessibility
- Interactive dashboard
- Easy sharing
- Browser compatibility



8.6 Flask Web Application

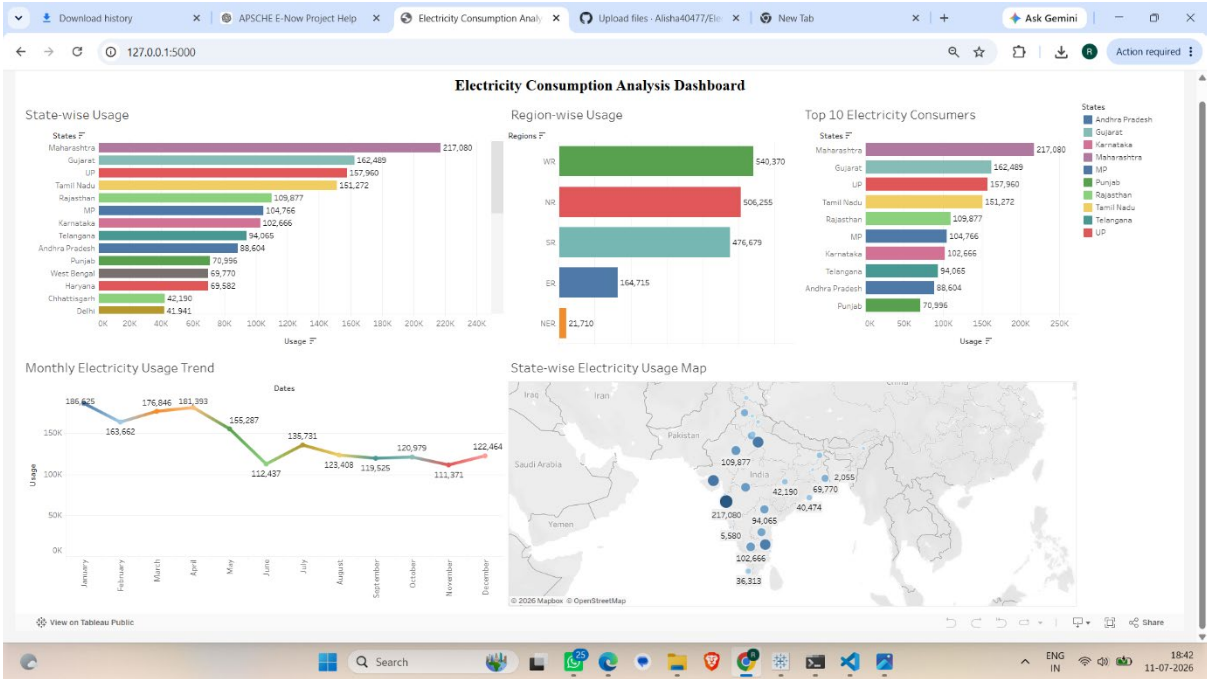
A Flask web application was developed to embed the Tableau Public dashboard.

Features

- Dashboard integration
- Simple navigation
- Browser-based access
- Local deployment

Application URL:

http://127.0.0.1:5000



8.7 GitHub Repository

The complete project was uploaded to GitHub.

Repository includes:

- Dataset
- SQL Queries
- Tableau Workbook
- Dashboard
- Story
- Flask Application
- Reports
- Screenshots
- README
- Requirements File

GitHub provides version control and project documentation.

Business Insights

The dashboard generated the following business insights:

- Electricity consumption varies significantly across different states.

- Regional electricity demand differs due to population and industrial activities.
- Monthly electricity consumption shows seasonal fluctuations.
- Some states consistently have higher electricity usage than others.
- Geographic maps help identify high-demand regions quickly.
- Interactive dashboards improve understanding of electricity consumption patterns.
- Business intelligence techniques simplify complex electricity data analysis.

Result Summary

Component	Status
Dataset Imported	Successfully Completed
MySQL Database	Successfully Created
SQL Queries	Successfully Executed
Tableau Worksheets	Successfully Developed
Dashboard	Successfully Created
Tableau Story	Successfully Developed
Tableau Public	Successfully Published
Flask Application	Successfully Integrated
GitHub Repository	Successfully Uploaded

Conclusion

The results demonstrate that the **Electricity Consumption Analysis** project successfully transformed raw electricity data into meaningful business insights. Interactive dashboards, maps, and stories enabled efficient analysis of electricity consumption across different states and regions. The integration of MySQL, Tableau, Tableau Public, and Flask provided an effective Business Intelligence solution that supports informed decision-making and energy planning.

---

9. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive and user-friendly dashboards.
- Accurate SQL-based analysis.
- Easy comparison between states and regions.
- Geographic visualization using maps.
- Monthly trend analysis.
- Business insights for decision-making.
- Web-based dashboard access.
- Faster report generation.
- Better electricity planning.
- Scalable architecture.

Disadvantages

- Depends on the quality of the dataset.
- Requires periodic data updates.
- Performance may decrease with extremely large datasets.
- Internet connection is required for Tableau Public.
- Limited to historical data analysis.
- Does not include real-time electricity monitoring.

---

10. CONCLUSION

The **Electricity Consumption Analysis** project successfully achieved its objective of developing an interactive Business Intelligence solution for analyzing electricity consumption patterns across different states and regions. By integrating **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, and **Flask**, the project provides an efficient platform for data storage, analysis, visualization, and web-based access.

The system enables users to perform state-wise and region-wise comparisons, analyze monthly electricity consumption trends, identify high electricity-consuming states, and generate meaningful business insights through interactive dashboards and Tableau Stories. The use of dynamic charts, maps, and filters simplifies data analysis and enhances the overall user experience.

The project demonstrates the practical application of Business Intelligence tools in transforming raw electricity consumption data into valuable information that supports informed decision-making. The successful implementation of SQL analysis, Tableau dashboards, Tableau Public publishing, and Flask integration confirms the reliability and effectiveness of the proposed solution.

Performance testing verified that the system is accurate, responsive, user-friendly, and compatible across the required software platforms. The project meets all the functional and non-functional requirements identified during the planning and design phases while providing a scalable foundation for future enhancements.

In conclusion, the **Electricity Consumption Analysis** project serves as an effective decision-support system for electricity boards, government organizations, researchers, and energy analysts. It improves the understanding of electricity consumption patterns, supports better resource planning, and demonstrates how modern Business Intelligence technologies can be used to solve real-world analytical problems efficiently.

11. FUTURE SCOPE

The Electricity Consumption Analysis project provides a strong foundation for future enhancements and can be extended by integrating advanced technologies and analytical capabilities. While the current system effectively analyzes historical electricity consumption data through interactive dashboards, several improvements can further increase its functionality and practical value.

Future enhancements may include:

- Integration with real-time electricity consumption data.
- AI and Machine Learning models for electricity demand forecasting.
- IoT-based smart meter integration for live monitoring.
- District-wise and city-wise electricity consumption analysis.
- Renewable energy production and consumption analysis.
- Cloud-based deployment for improved scalability and accessibility.
- Mobile application support for anytime, anywhere access.
- Automated report generation and scheduled dashboard updates.
- Predictive analytics for resource planning and energy optimization.
- Smart grid analytics for efficient electricity distribution and management.

These enhancements will improve the system's scalability, accuracy, and analytical capabilities while supporting more effective energy management, policy planning, and data-driven decision-making in the future.

.

---

12. APPENDIX

The Appendix contains the supporting resources, project artifacts, and reference materials used during the development of the Electricity Consumption Analysis project. These resources provide access to the dataset, source code, dashboards, application, and technical tools used for implementation and documentation.

12.1 Project Resources

Resource	Details
Project Title	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Dataset	Electricity Consumption Dataset (CSV Format)
Database	MySQL Database – electricity
Table Name	consumption
Programming Language	Python 3.12.7
Database Tool	MySQL Workbench
Visualization Tool	Tableau Desktop
Dashboard Publishing	Tableau Public
Web Framework	Flask
IDE	Visual Studio Code
Version Control	Git & GitHub
Documentation	Microsoft Word, PDF

12.2 Project Links

Dataset

https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view

GitHub Repository

https://github.com/Alisha40477/Electricity-Consumption-Analysis

Tableau Public Dashboard

https://public.tableau.com/app/profile/mohammed.mahaboob.alisha.shaik/viz/Electricity_Consumption_Analysis_17837707990200/Dashboard1?publish=yes

Flask Web Application (Local)

http://127.0.0.1:5000

12.3 Project Deliverables

The following deliverables were completed during the project:

- Electricity Consumption Dataset
- MySQL Database (electricity)
- SQL Query File (queries.sql)
- Tableau Workbook (.twb)
- Interactive Tableau Dashboard
- Tableau Story
- Tableau Public Dashboard
- Flask Web Application
- GitHub Repository
- Project Documentation
- Final Project Report

12.4 References

The following resources were referred to during the development of the project:

1. Tableau Desktop Documentation
2. Tableau Public Documentation
3. MySQL Official Documentation
4. Flask Official Documentation
5. Python Official Documentation
6. GitHub Documentation
7. Electricity Consumption Dataset

12.5 Declaration

This project was developed as part of the APSCHE Data Analytics Internship. All project activities, including database development, SQL analysis, dashboard creation, web application integration, testing, and documentation, were completed using the tools and technologies mentioned in this report. The project is intended for educational and academic purposes.